

52 kHz Simple 3A Buck Regulator

Features

- 3.3V, 5V, 12V, and Adjustable Output Versions
- Voltage over Specified Line and Load Conditions:
 - Fixed Version: $\pm 3\%$ Max. Output Voltage
 - Adjustable Version: $\pm 2\%$ Max. Feedback Voltage
- Specified 3A Output Current
- Wide Input Voltage Range of 4V to 40V
- Wide Output Voltage Range of 1.23V to 37V
- Requires Only Four External Components
- 52 kHz Fixed-Frequency Internal Oscillator
- Low Power Standby Mode I_Q Typically $< 200 \mu A$
- 80% Efficiency (Adjustable Version Typically $> 80\%$)
- Uses Readily Available Standard Inductors
- Thermal Shutdown and Current Limit Protection
- 100% Electrical Thermal Limit Built-In

Applications

- Simple High-Efficiency Step-Down (Buck) Regulator
- Efficient Pre-Regulator for Linear Regulators
- On-Card Switching Regulators
- Positive and Negative Converter (Inverting Buck-Boost)
- Isolated Flyback Converter using Minimum Number of External Components
- Negative Boost Converter

General Description

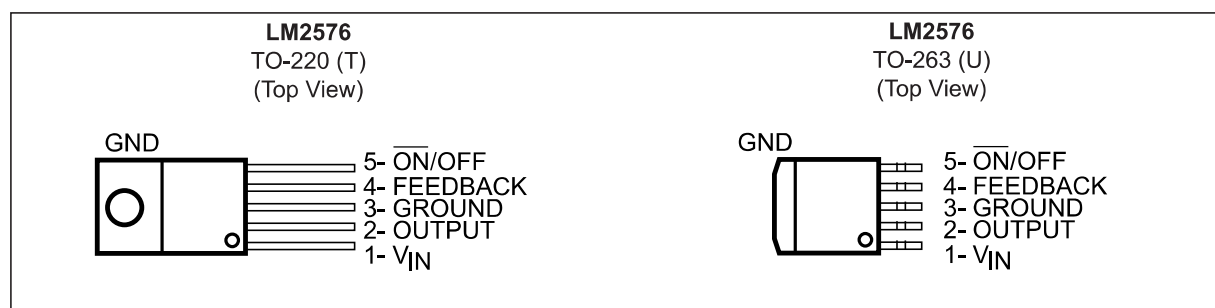
The LM2576 series of monolithic integrated circuits provide all the active functions for a step-down (buck) switching regulator. Fixed versions are available with a 3.3V, 5V, or 12V fixed output. Adjustable versions have an output voltage range from 1.23V to 37V. Both versions are capable of driving a 3A load with excellent line and load regulation.

These regulators are simple to use because they require a minimum number of external components and include internal frequency compensation and a fixed-frequency oscillator. The LM2576 series offers a high efficiency replacement for popular three-terminal adjustable linear regulators. It substantially reduces the size of the heat sink, and in many cases no heat sink is required.

A standard series of inductors available from several different manufacturers are ideal for use with the LM2576 series. This feature greatly simplifies the design of switch-mode power supplies.

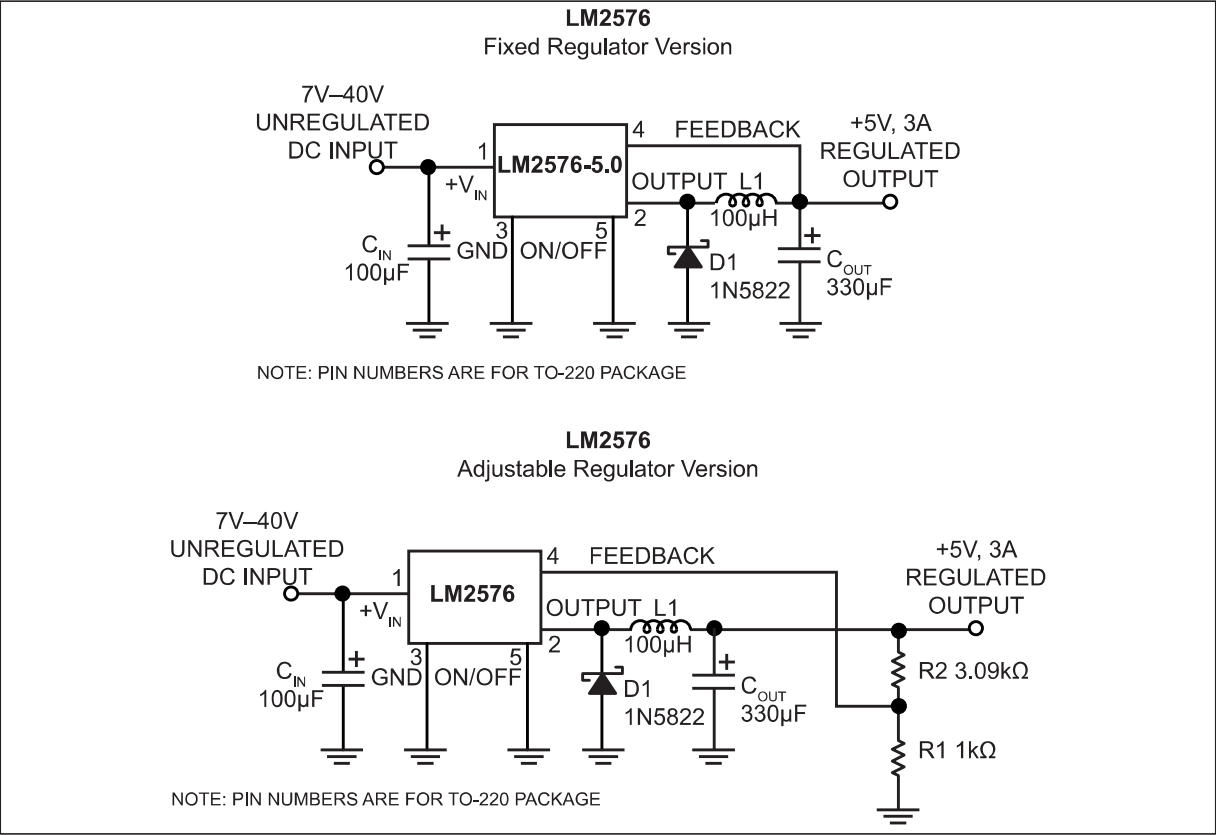
The feedback voltage is guaranteed to $\pm 2\%$ tolerance for adjustable versions, and the output voltage is guaranteed to $\pm 3\%$ for fixed versions, within specified input voltages and output load conditions. The oscillator frequency is guaranteed to $\pm 10\%$. External shutdown is included, featuring less than $200 \mu A$ standby current. The output switch includes cycle-by-cycle current limiting and thermal shutdown for full protection under fault conditions.

Package Types

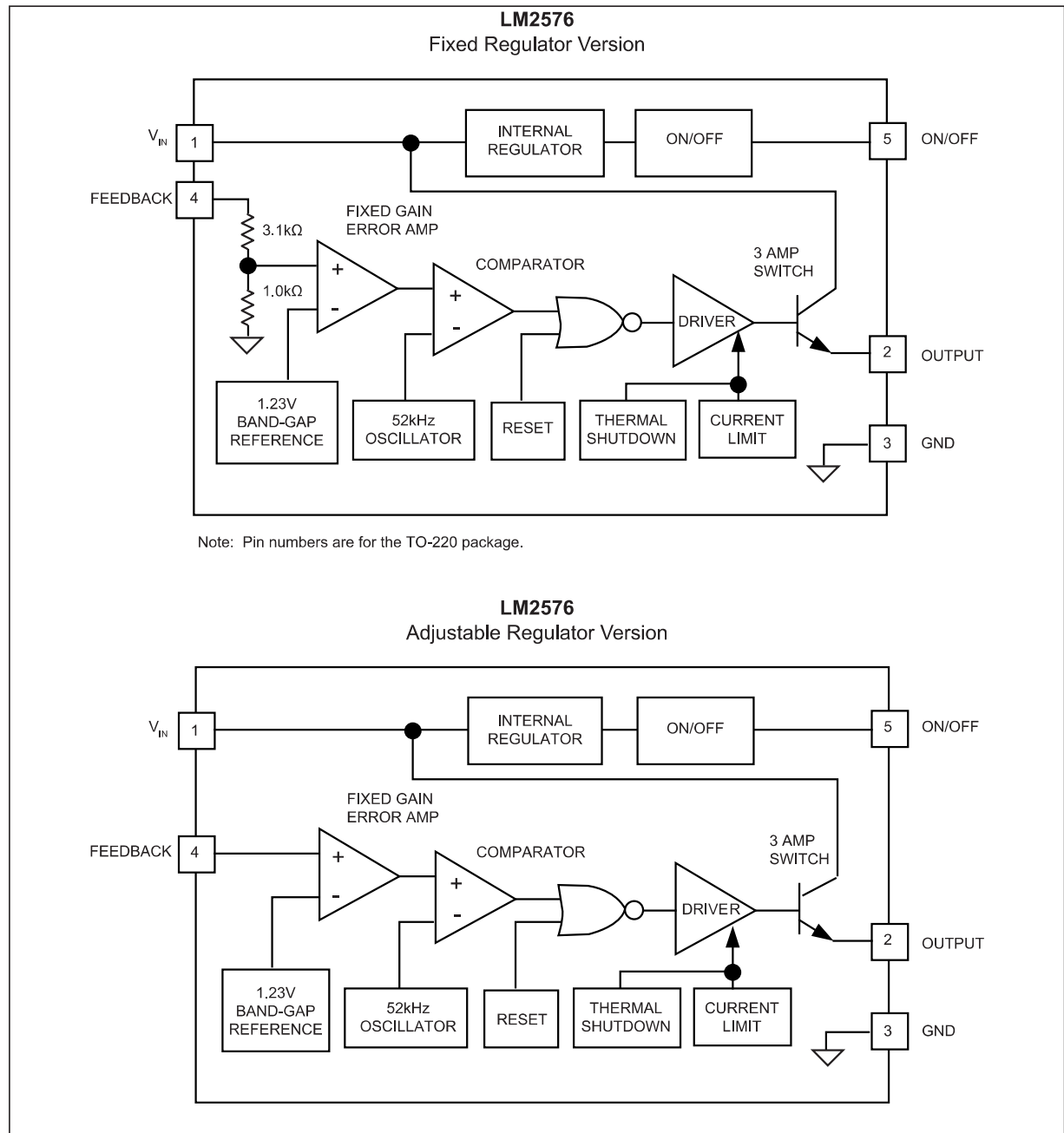


LM2576

Typical Application Circuits



Functional Block Diagrams



LM2576

1.0 ELECTRICAL CHARACTERISTICS

Absolute Maximum Ratings †

Maximum Supply Voltage.....	+45V
ON/OFF Pin Input Voltage.....	$-0.3V \leq V \leq +40V$
Output Voltage to Ground.....	-1V
Power Dissipation.....	Internally Limited
ESD Rating (Note 1)	
C = 100 pF, R = 1.5 kΩ.....	2 kV
FB Pin.....	1 kV

Operating Ratings ††

Supply Voltage (V_{IN}).....	+40V
----------------------------------	------

† **Notice:** Stresses above those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operational sections of this specification is not intended. Exposure to maximum rating conditions for extended periods may affect device reliability.

†† **Notice:** The device is not guaranteed to function outside its operating ratings.

Note 1: Devices are ESD sensitive. Handling precautions recommended. Human body model, 1.5 kΩ in series with 100 pF.

ELECTRICAL CHARACTERISTICS

Specifications with standard typeface are for $T_J = +25^\circ\text{C}$, **bold** values are valid for $-40^\circ\text{C} \leq T_J \leq +125^\circ\text{C}$. Unless otherwise specified, $V_{IN} = 12V$, and $I_{LOAD} = 500\text{ mA}$. [Note 1](#)

Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
System Parameters, Adjustable Regulators (Note 2) Test Circuit Figure 4-1						
Feedback Voltage	V_{OUT}	1.217	1.230	1.243	V	$V_{IN} = 12V$, $I_{LOAD} = 0.5A$, $V_{OUT} = 5V$
Feedback Voltage LM2576	V_{OUT}	1.193	1.230	1.267	V	$0.5A \leq I_{LOAD} \leq 3A$, $8V \leq V_{IN} \leq 40V$, $V_{OUT} = 5V$
		1.180	—	1.280		
Efficiency	η	—	82	—	%	$V_{IN} = 12V$, $I_{LOAD} = 3A$, $V_{OUT} = 5V$
System Parameters, 3.3V Regulators (Note 2) Test Circuit Figure 4-1						
Output Voltage	V_{OUT}	3.234	3.3	3.363	V	$V_{IN} = 12V$, $I_{LOAD} = 0.5A$, $V_{OUT} = 3.3V$
Output voltage LM2576-3.3	V_{OUT}	3.168	3.3	3.432	V	$0.5A \leq I_{LOAD} \leq 3A$, $6V \leq V_{IN} \leq 40V$, $V_{OUT} = 3.3V$
		3.135	—	3.465		
Efficiency	η	—	75	—	%	$V_{IN} = 12V$, $I_{LOAD} = 3A$
System Parameters, 5V Regulators (Note 2) Test Circuit Figure 4-1						
Output Voltage	V_{OUT}	4.900	5.0	5.100	V	$V_{IN} = 12V$, $I_{LOAD} = 0.5A$, $V_{OUT} = 5.0V$
Output voltage LM2576-5.0	V_{OUT}	4.800	5.0	5.200	V	$0.5A \leq I_{LOAD} \leq 3A$, $8V \leq V_{IN} \leq 40V$, $V_{OUT} = 5.0V$
		4.750	—	5.250		

Note 1: Specification for packaged product only.

- External components such as the catch diode, inductor, input and output capacitors can affect switching regulator system performance. When the LM2576/LM1576 is used as shown in [Figure 4-1](#) test circuit, system performance will be shown in system parameters section of Electrical Characteristics.
- Output (pin 2) sourcing current. No diode, inductor or capacitor connected to output.
- Feedback (pin 4) removed from output and connected to 0V.
- Feedback (pin 4) removed from output and connected to 12V to force the output transistor OFF.

ELECTRICAL CHARACTERISTICS (CONTINUED)

Specifications with standard typeface are for $T_J = +25^\circ\text{C}$, **bold** values are valid for $-40^\circ\text{C} \leq T_J \leq +125^\circ\text{C}$. Unless otherwise specified, $V_{IN} = 12\text{V}$, and $I_{LOAD} = 500\text{ mA}$. [Note 1](#)

Parameter	Sym.	Min.	Typ.	Max.	Units	Conditions
Efficiency	η	—	82	—	%	$V_{IN} = 12\text{V}$, $I_{LOAD} = 3\text{A}$
System Parameters, 12V Regulators (Note 2) Test Circuit Figure 4-1						
Output Voltage	V_{OUT}	11.760	12.0	12.240	V	$V_{IN} = 25\text{V}$, $I_{LOAD} = 0.5\text{A}$, $V_{OUT} = 12\text{V}$
Output voltage LM2576-12	V_{OUT}	11.520 11.400	12.0 —	12.480 12.600	V	$0.5\text{A} \leq I_{LOAD} \leq 3\text{A}$, $15\text{V} \leq V_{IN} \leq 40\text{V}$, $V_{OUT} = 12\text{V}$
Efficiency	η	—	88	—	%	$V_{IN} = 25\text{V}$, $I_{LOAD} = 3\text{A}$
Device Parameters, Adjustable Regulator						
Feedback Bias Current	I_B	100	50	500	nA	$V_{OUT} = 5\text{V}$
Device Parameters, Fixed and Adjustable Regulators						
Oscillator Frequency	f_O	47 42	52 —	58 63	kHz	—
Saturation Voltage	V_{SAT}	— —	1.4 —	1.8 2.0	V	$I_{OUT} = 3\text{A}$, Note 3
Max. Duty Cycle (ON)	DC	93	98	—	%	Note 4
Current Limit	I_{LIM}	4.2 3.5	5.8 —	6.9 7.5	A	Peak current, $t_{ON} \leq 3\text{ }\mu\text{s}$, Note 3
Output Leakage Current	I_L	—	—	2	mA	$V_{IN} = 40\text{V}$, Note 5 , Output = 0V
		—	7.5	—		Output = -1V
		—	—	30		Note 5 , Output = -1V
Quiescent Current	I_Q	—	5	10	mA	Note 5
Standby Quiescent Current	I_{STBY}	—	50	200	μA	ON/OFF Pin = 5V (OFF)
On/Off Control, Fixed and Adjustable Regulators , Test Circuit Figure 4-1						
ON/OFF Pin Logic Input Level	V_{IH}	2.2	1.4	2.4	V	$V_{OUT} = 0\text{V}$
	V_{IL}	1.0	1.2	0.8		$V_{OUT} = 5\text{V}$
ON/OFF Pin Logic Current	I_{IH}	—	4	30	μA	ON/OFF Pin = 5V (OFF)
	I_{IL}	—	0.01	10		ON/OFF Pin = 0V (ON)

Note 1: Specification for packaged product only.

2: External components such as the catch diode, inductor, input and output capacitors can affect switching regulator system performance. When the LM2576/LM1576 is used as shown in [Figure 4-1](#) test circuit, system performance will be shown in system parameters section of Electrical Characteristics.

3: Output (pin 2) sourcing current. No diode, inductor or capacitor connected to output.

4: Feedback (pin 4) removed from output and connected to 0V.

5: Feedback (pin 4) removed from output and connected to 12V to force the output transistor OFF.

LM2576

TEMPERATURE SPECIFICATIONS

Parameters	Sym.	Min.	Typ.	Max.	Units	Conditions
Temperature Ranges						
Maximum Junction Temperature	T_J	—	—	+150	°C	—
Storage Temperature Range	T_S	–65	—	+150	°C	—
Lead Temperature	—	—	—	+260	°C	Soldering, 10 sec.
Operating Temperature Range	T_A	–40	—	+125	°C	—
Package Thermal Resistance						
Thermal Resistance, TO-220, TO-263	θ_{JA}	—	65	—	°C/W	Junction-to-Ambient, Note 2
Thermal Resistance, TO-220, TO-263	θ_{JA}	—	45	—	°C/W	Junction-to-Ambient, Note 3
Thermal Resistance, TO-220, TO-263	θ_{JC}	—	2	—	°C/W	Junction-to-Case

- Note 1:** The maximum allowable power dissipation is a function of ambient temperature, the maximum allowable junction temperature and the thermal resistance from junction to air (i.e., T_A , T_J , θ_{JA}). Exceeding the maximum allowable power dissipation will cause the device operating junction temperature to exceed the maximum +125°C rating. Sustained junction temperatures above +125°C can impact the device reliability.
- 2:** Junction to ambient thermal resistance (no external heat sink) for the 5-lead TO-220 package mounted vertically, with 1/2" leads in a socket, or on PC board with minimum copper area.
- 3:** Junction to ambient thermal resistance (no external heat sink) for the 5-lead TO-220 package mounted vertically, with 1/4" leads soldered to PC board containing approximately 4 square inches of copper area surrounding the leads.
- 4:** Junction to ambient thermal resistance with approximately 1 square inch of PC board copper surrounding the leads. Additional copper will lower thermal resistance further.

2.0 TYPICAL PERFORMANCE CURVES

Note: The graphs and tables provided following this note are a statistical summary based on a limited number of samples and are provided for informational purposes only. The performance characteristics listed herein are not tested or guaranteed. In some graphs or tables, the data presented may be outside the specified operating range (e.g., outside specified power supply range) and therefore outside the warranted range.

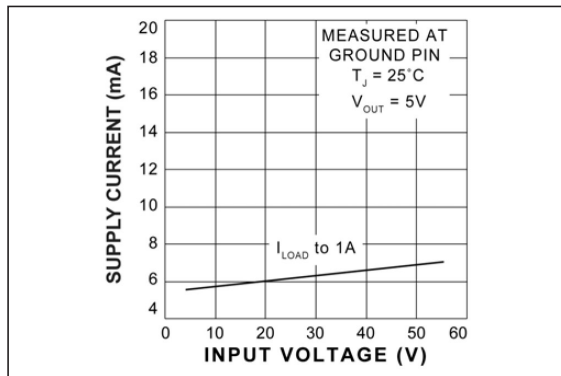


FIGURE 2-1: Supply Current.

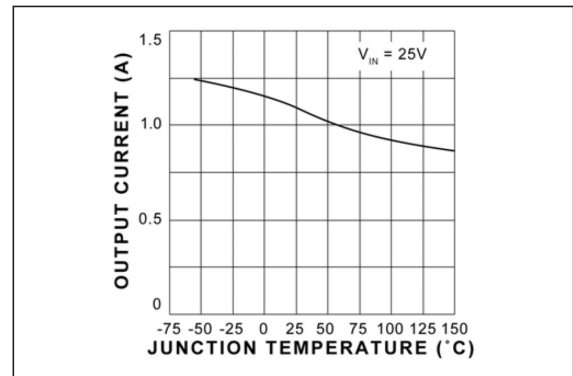


FIGURE 2-4: Current Limit.

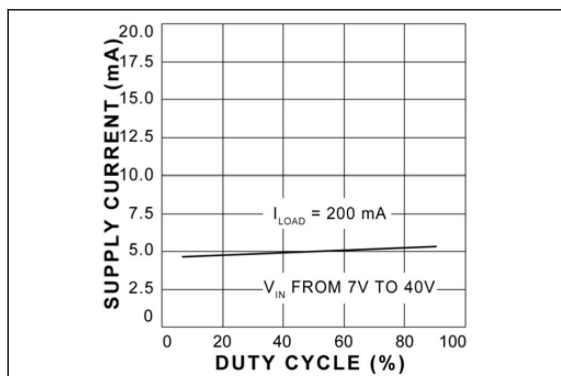


FIGURE 2-2: Supply Current vs. Duty Cycle.

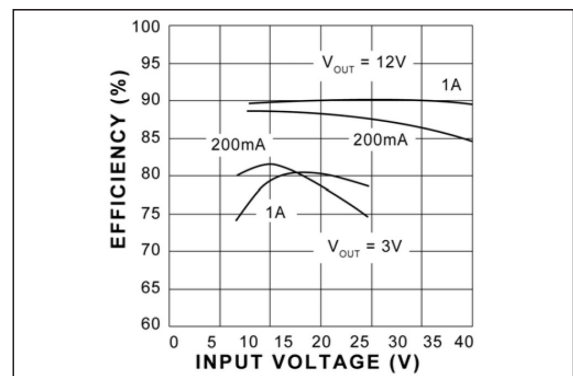


FIGURE 2-5: Efficiency.

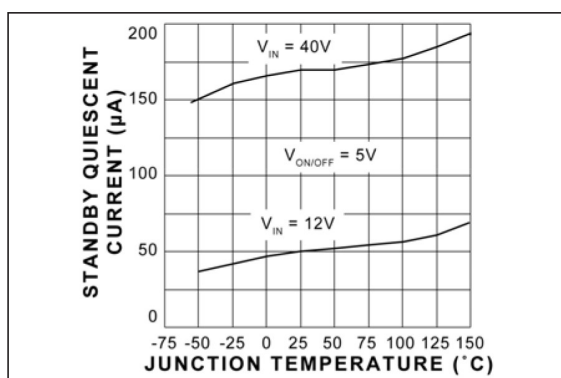


FIGURE 2-3: Standby Quiescent Current.

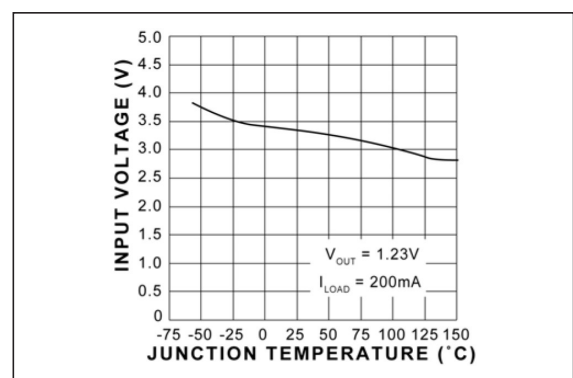


FIGURE 2-6: Minimum Operating Voltage.

LM2576

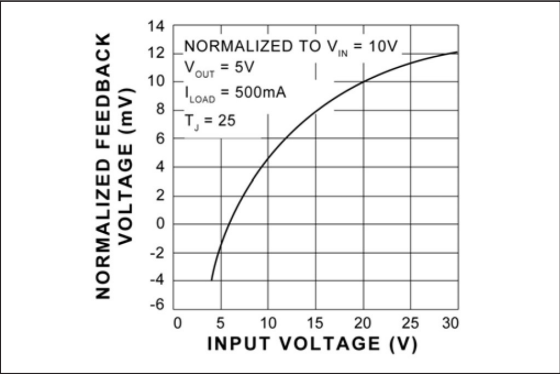


FIGURE 2-7: Line Regulation.

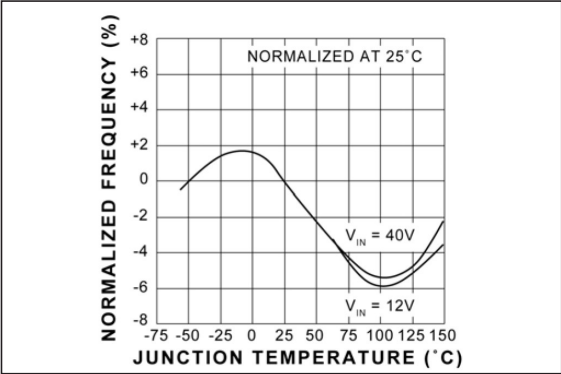


FIGURE 2-10: Oscillator Frequency.

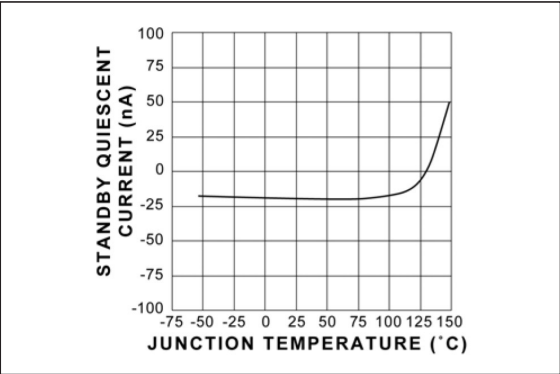


FIGURE 2-8: Feedback Pin Current.

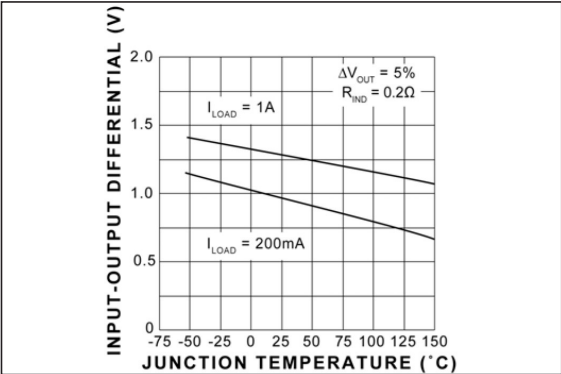


FIGURE 2-11: Dropout Voltage.

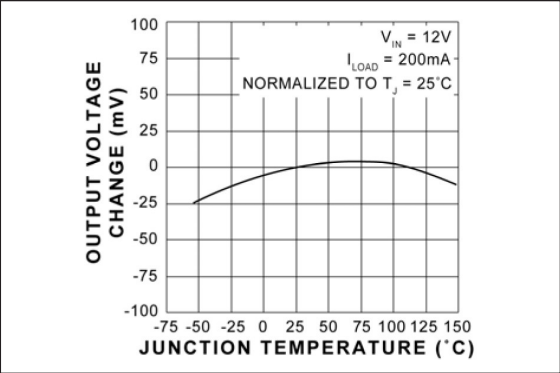


FIGURE 2-9: Normalized Output Voltage.

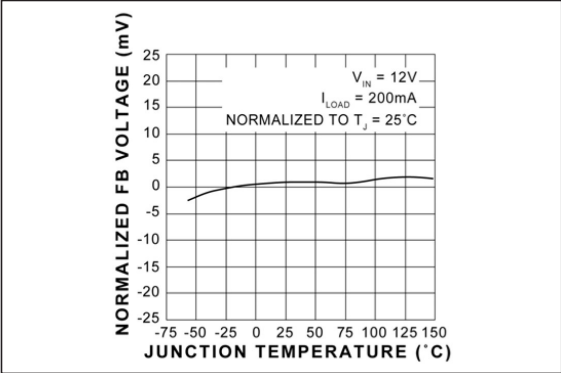


FIGURE 2-12: Normalized Feedback Voltage (Adjustable Version Only).

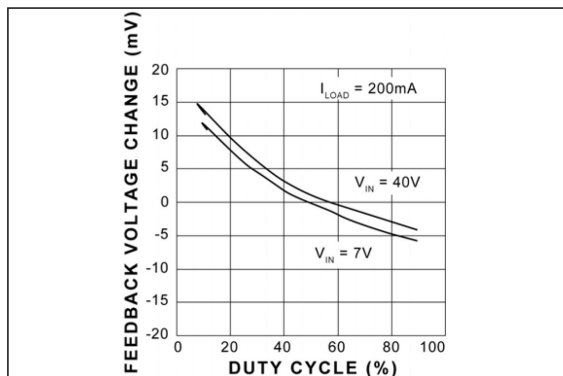


FIGURE 2-13: Feedback Voltage vs. Duty Cycle (Adjustable Version Only).

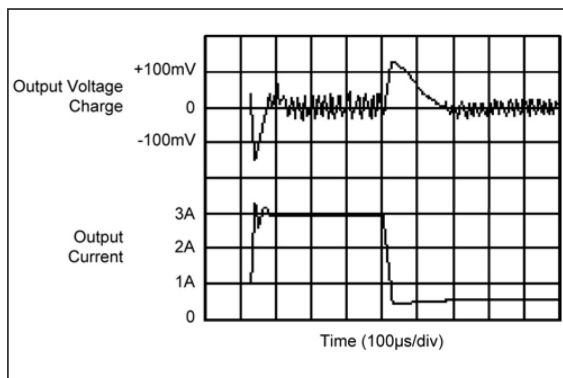


FIGURE 2-14: Load Transient Response.

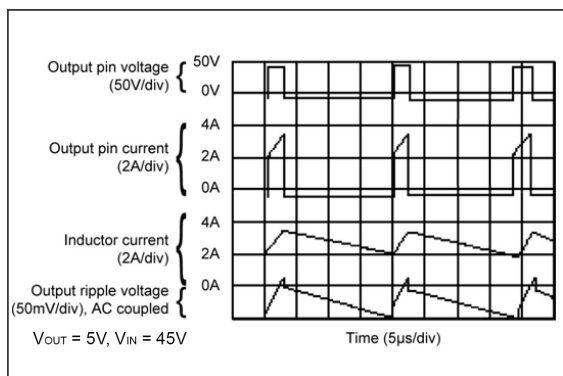


FIGURE 2-15: Switching Waveforms.

LM2576

3.0 PIN DESCRIPTIONS

The descriptions of the pins are listed in [Table 3-1](#).

TABLE 3-1: PIN FUNCTION TABLE

Pin Number	Pin Name	Description
1	V _{IN}	Supply input. Requires bypass capacitor to GND.
2	OUTPUT	Switch output. Internal MOSFET switch output.
3	GND	Ground.
4	FB	Feedback. For fixed output versions, connect to the output. For adjustable versions, connect to external resistive divider to set output voltage.
5	ON/OFF	Enable. Logic low enables operation. Logic high shuts down the regulator. Do not leave floating.

4.0 TEST CIRCUIT

As in any switching regulator, layout is very important. Rapidly switching currents associated with wiring inductance generate voltage transients which can cause problems. For minimal stray inductance and ground loops, the length of the leads indicated by heavy lines should be kept as short as possible. Single point grounding (as indicated) or grounding plane construction should be used for best results.

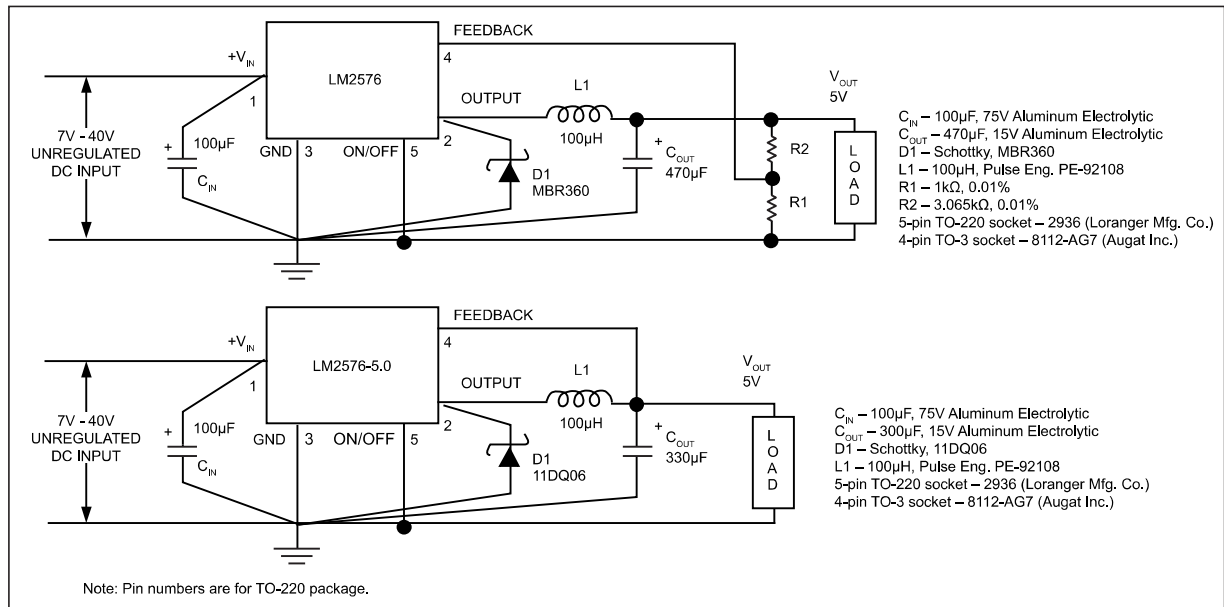


FIGURE 4-1: Test Circuit.

LM2576

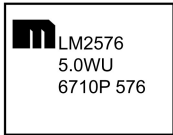
5.0 PACKAGING INFORMATION

5.1 Package Marking Information

5-Lead TO-220*
5-Lead TO-263*
(Fixed Versions)



Example



5-Lead TO-220*
5-Lead TO-263*
(Adj. Versions)



Example



Legend:	XX...X	Product code or customer-specific information
	Y	Year code (last digit of calendar year)
	YY	Year code (last 2 digits of calendar year)
	WW	Week code (week of January 1 is week '01')
	NNN	Alphanumeric traceability code
	e3	Pb-free JEDEC® designator for Matte Tin (Sn)
	*	This package is Pb-free. The Pb-free JEDEC designator (e3) can be found on the outer packaging for this package.
	•, ▲, ▼	Pin one index is identified by a dot, delta up, or delta down (triangle mark).

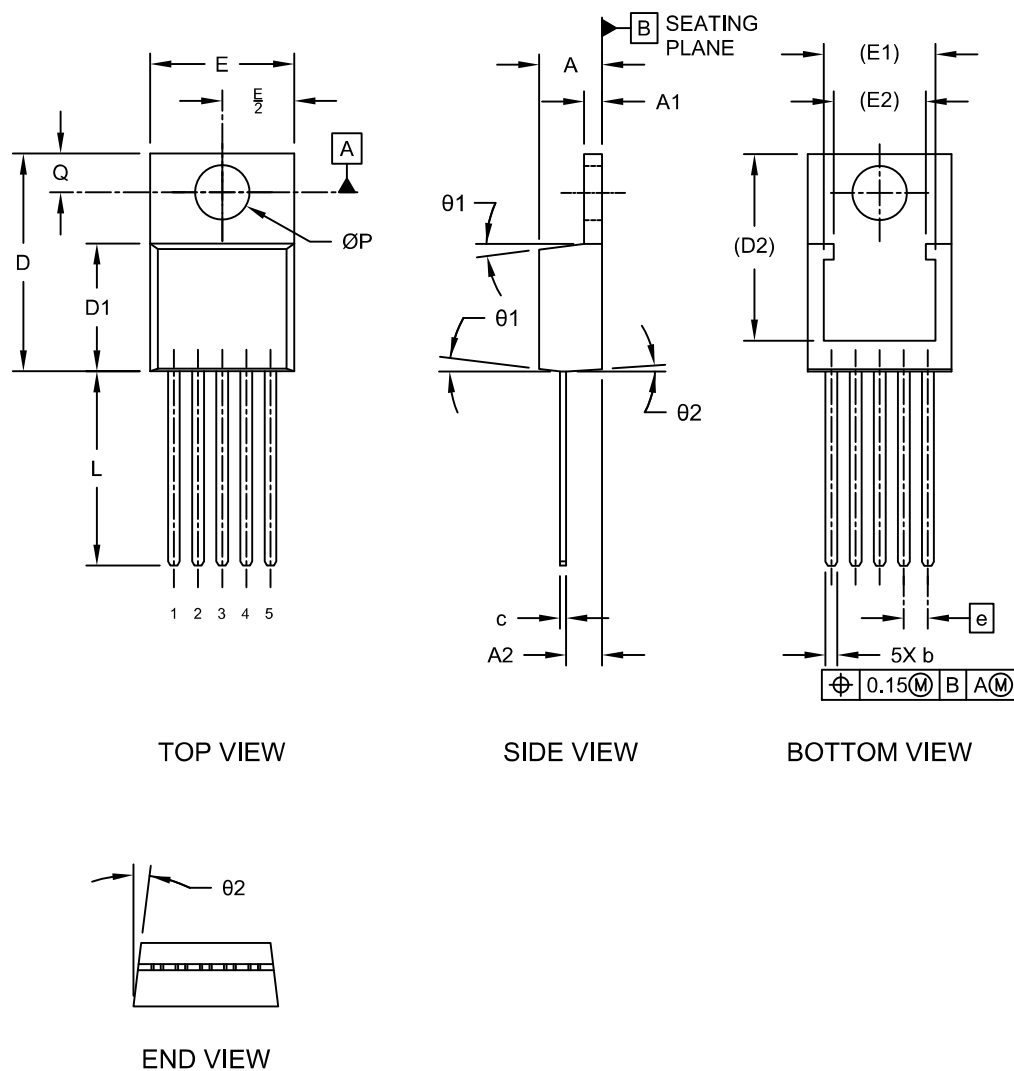
Note: In the event the full Microchip part number cannot be marked on one line, it will be carried over to the next line, thus limiting the number of available characters for customer-specific information. Package may or may not include the corporate logo.

Underbar (¯) and/or Overbar (¯) symbol may not be to scale.

5-Lead TO-220 Package Outline and Recommended Land Pattern

5-Lead Transistor Outline Type LB03 (B8X) - [TO-220] Micrel Legacy Package TO220-LB03-5LD-PL-1

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

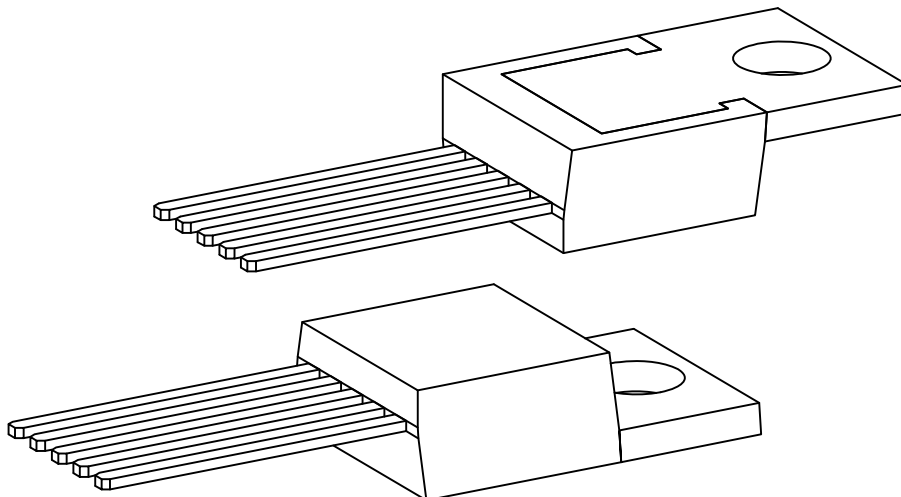


Microchip Technology Drawing C04-036 Rev C Sheet 1 of 2

LM2576

5-Lead Transistor Outline Type LB03 (B8X) - [TO-220] Micrel Legacy Package TO220-LB03-5LD-PL-1

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits		INCHES		
		Min	Nom	Max
Number of Leads	N	5		
Pitch	e	.067 BSC		
Overall Height	A	.160	.175	.190
Tab Height	A1	.045	.050	.055
Seating Plane to Lead	A2	.080	.098	.115
Lead Width	b	.025	.033	.040
Lead Thickness	c	.012	.016	.020
Lead Length	L	.500	.540	.580
Total Body Length Including Tab	D	.542	.580	.619
Molded Body Length	D1	.348	.354	.360
Total Width	E	.380	.400	.420
Pad Width	E1	0.256 REF		
Pad Length	D2	0.486 REF		
Hole Diameter	ØP	.146	.151	.156
Hole Center to Tab Edge	Q	.103	.108	.113
Molded Body Draft Angle	Ø2	3	7	10
Molded Body Draft Angle	Ø2	1	4	7

Notes:

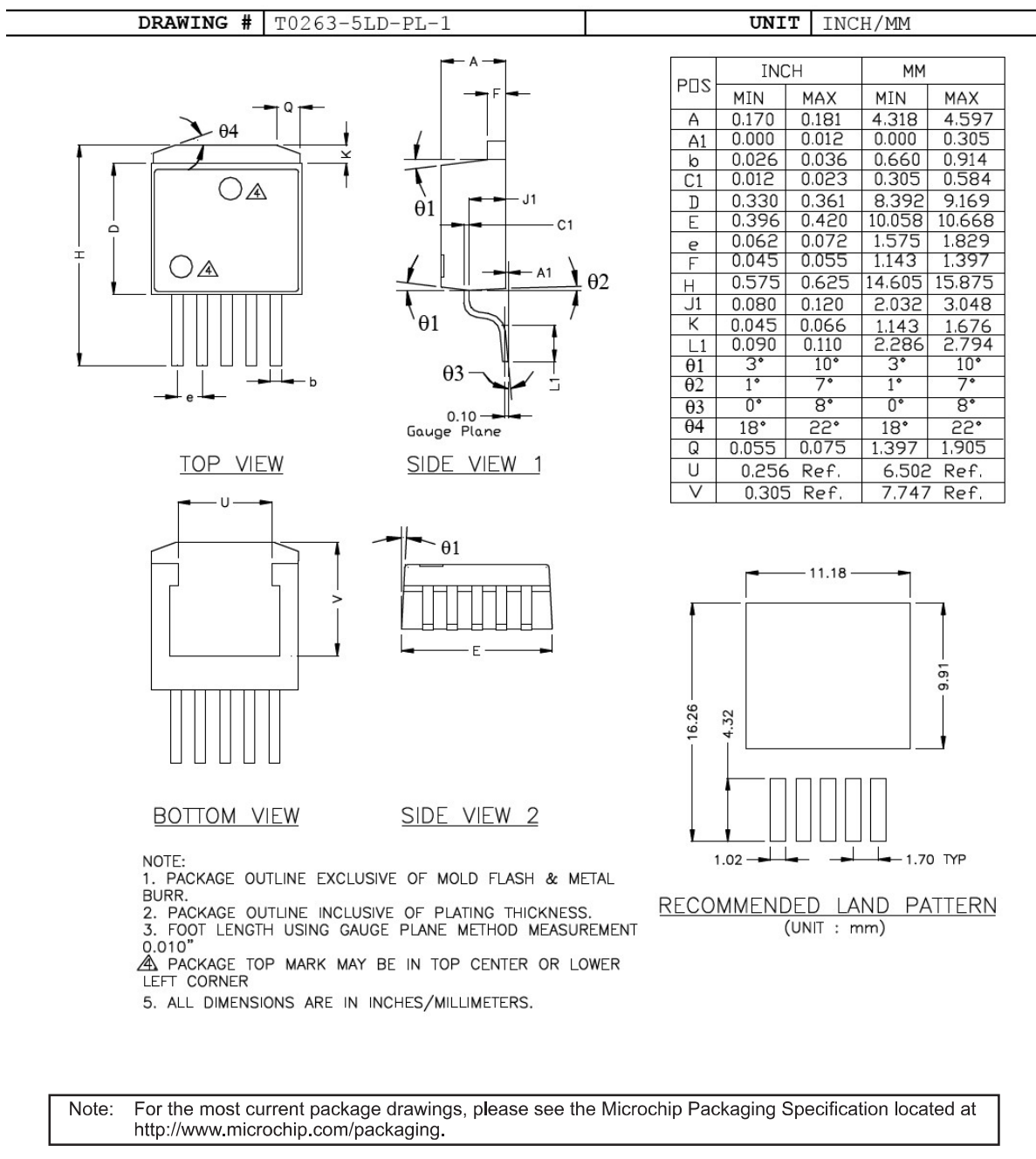
- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-036 Rev C Sheet 2 of 2

5-Lead TO-263 Package Outline and Recommended Land Pattern

TITLE

5 LEAD T0263 PACKAGE OUTLINE & RECOMMENDED LAND PATTERN



LM2576

NOTES:

APPENDIX A: REVISION HISTORY

Revision A (August 2019)

- Converted Micrel document LM2576 to Microchip data sheet template DS20006238A.
- Minor grammatical text changes throughout.

LM2576

NOTES:

PRODUCT IDENTIFICATION SYSTEM

To order or obtain information, e.g., on pricing or delivery, contact your local Microchip representative or sales office.

Device	-XX	X	X	-XX
Part No.	Output Voltage	Junction Temp. Range	Package	Media Type
Device:	LM2576:	52 kHz Simple 3A Buck Regulator		
Output Voltage:	<blank> = Adjustable 3.3 = 3.3V 5.0 = 5.0V 12 = 12V			
Junction Temperature Range:	W = -40°C to +125°C, RoHS-Compliant			
Package:	T = 5-Lead TO-220 U = 5-Lead TO-263			
Media Type:	<blank>= 50/Tube TR = 750/Reel			

Examples:	
a) LM2576WT:	LM2576, Adj. Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-220, 50/Tube
b) LM2576-3.3WT:	LM2576, 3.3V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-220, 50/Tube
c) LM2576-5.0WT:	LM2576, 5.0V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-220, 50/Tube
d) LM2576-12WT:	LM2576, 12V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-220, 50/Tube
e) LM2576WU:	LM2576, Adj. Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 50/Tube
f) LM2576-3.3WU:	LM2576, 3.3V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 50/Tube
g) LM2576-5.0WU:	LM2576, 5.0V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 50/Tube
h) LM2576-12WU:	LM2576, 12V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 50/Tube
i) LM2576WU-TR:	LM2576, Adj. Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 750/Reel
j) LM2576-3.3WU-TR:	LM2576, 3.3V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 750/Reel
k) LM2576-5.0WU-TR:	LM2576, 5.0V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 750/Reel
l) LM2576-12WU-TR:	LM2576, 12V Output Voltage, -40°C to +125°C Temp. Range, 5-Lead TO-263, 750/Reel
Note 1:	Tape and Reel identifier only appears in the catalog part number description. This identifier is used for ordering purposes and is not printed on the device package. Check with your Microchip Sales Office for package availability with the Tape and Reel option.

LM2576

NOTES:

Note the following details of the code protection feature on Microchip devices:

- Microchip products meet the specification contained in their particular Microchip Data Sheet.
- Microchip believes that its family of products is one of the most secure families of its kind on the market today, when used in the intended manner and under normal conditions.
- There are dishonest and possibly illegal methods used to breach the code protection feature. All of these methods, to our knowledge, require using the Microchip products in a manner outside the operating specifications contained in Microchip's Data Sheets. Most likely, the person doing so is engaged in theft of intellectual property.
- Microchip is willing to work with the customer who is concerned about the integrity of their code.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of their code. Code protection does not mean that we are guaranteeing the product as "unbreakable."

Code protection is constantly evolving. We at Microchip are committed to continuously improving the code protection features of our products. Attempts to break Microchip's code protection feature may be a violation of the Digital Millennium Copyright Act. If such acts allow unauthorized access to your software or other copyrighted work, you may have a right to sue for relief under that Act.

Information contained in this publication regarding device applications and the like is provided only for your convenience and may be superseded by updates. It is your responsibility to ensure that your application meets with your specifications. MICROCHIP MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WHETHER EXPRESS OR IMPLIED, WRITTEN OR ORAL, STATUTORY OR OTHERWISE, RELATED TO THE INFORMATION, INCLUDING BUT NOT LIMITED TO ITS CONDITION, QUALITY, PERFORMANCE, MERCHANTABILITY OR FITNESS FOR PURPOSE. Microchip disclaims all liability arising from this information and its use. Use of Microchip devices in life support and/or safety applications is entirely at the buyer's risk, and the buyer agrees to defend, indemnify and hold harmless Microchip from any and all damages, claims, suits, or expenses resulting from such use. No licenses are conveyed, implicitly or otherwise, under any Microchip intellectual property rights unless otherwise stated.

Trademarks

The Microchip name and logo, the Microchip logo, Adaptec, AnyRate, AVR, AVR logo, AVR Freaks, BesTime, BitCloud, chipKIT, chipKIT logo, CryptoMemory, CryptoRF, dsPIC, FlashFlex, flexPWR, HELDO, IGLOO, JukeBlox, KeeLoq, Kleer, LANCheck, LinkMD, maXStylus, maXTouch, MediaLB, megaAVR, Microsemi, Microsemi logo, MOST, MOST logo, MPLAB, OptoLyzer, PackTime, PIC, picoPower, PICSTART, PIC32 logo, PolarFire, Prochip Designer, QTouch, SAM-BA, SenGenuity, SpyNIC, SST, SST Logo, SuperFlash, Symmetricom, SyncServer, Tachyon, TempTrackr, TimeSource, tinyAVR, UNI/O, Vectron, and XMEGA are registered trademarks of Microchip Technology Incorporated in the U.S.A. and other countries.

APT, ClockWorks, The Embedded Control Solutions Company, EtherSynch, FlashTec, Hyper Speed Control, HyperLight Load, IntelliMOS, Libero, motorBench, mTouch, Powermite 3, Precision Edge, ProASIC, ProASIC Plus, ProASIC Plus logo, Quiet-Wire, SmartFusion, SyncWorld, Temux, TimeCesium, TimeHub, TempPictra, TimeProvider, Vite, WinPath, and ZL are registered trademarks of Microchip Technology Incorporated in the U.S.A.

Adjacent Key Suppression, AKS, Analog-for-the-Digital Age, Any Capacitor, AnyIn, AnyOut, BlueSky, BodyCom, CodeGuard, CryptoAuthentication, CryptoAutomotive, CryptoCompanion, CryptoController, dsPICDEM, dsPICDEM.net, Dynamic Average Matching, DAM, ECAN, EtherGREEN, In-Circuit Serial Programming, ICSP, INICnet, Inter-Chip Connectivity, JitterBlocker, KlearNet, KlearNet logo, memBrain, Mindi, MiWi, MPASM, MPF, MPLAB Certified logo, MPLIB, MPLINK, MultiTRAK, NetDetach, Omniscient Code Generation, PICDEM, PICDEM.net, PICkit, PICtail, PowerSmart, PureSilicon, QMatrix, REAL ICE, Ripple Blocker, SAM-ICE, Serial Quad I/O, SMART-I.S., SQI, SuperSwitcher, SuperSwitcher II, Total Endurance, TSHARC, USBCheck, VariSense, ViewSpan, WiperLock, Wireless DNA, and ZENA are trademarks of Microchip Technology Incorporated in the U.S.A. and other countries.

SQTP is a service mark of Microchip Technology Incorporated in the U.S.A.

The Adaptec logo, Frequency on Demand, Silicon Storage Technology, and Symmcom are registered trademarks of Microchip Technology Inc. in other countries.

GestIC is a registered trademark of Microchip Technology Germany II GmbH & Co. KG, a subsidiary of Microchip Technology Inc., in other countries.

All other trademarks mentioned herein are property of their respective companies.

© 2019, Microchip Technology Incorporated, All Rights Reserved.

ISBN: 978-1-5224-4901-0

For information regarding Microchip's Quality Management Systems, please visit www.microchip.com/quality.



Worldwide Sales and Service

AMERICAS

Corporate Office
2355 West Chandler Blvd.
Chandler, AZ 85224-6199
Tel: 480-792-7200
Fax: 480-792-7277
Technical Support:
[http://www.microchip.com/
support](http://www.microchip.com/support)
Web Address:
www.microchip.com

Atlanta
Duluth, GA
Tel: 678-957-9614
Fax: 678-957-1455

Austin, TX
Tel: 512-257-3370

Boston
Westborough, MA
Tel: 774-760-0087
Fax: 774-760-0088

Chicago
Itasca, IL
Tel: 630-285-0071
Fax: 630-285-0075

Dallas
Addison, TX
Tel: 972-818-7423
Fax: 972-818-2924

Detroit
Novi, MI
Tel: 248-848-4000

Houston, TX
Tel: 281-894-5983

Indianapolis
Noblesville, IN
Tel: 317-773-8323
Fax: 317-773-5453
Tel: 317-536-2380

Los Angeles
Mission Viejo, CA
Tel: 949-462-9523
Fax: 949-462-9608
Tel: 951-273-7800

Raleigh, NC
Tel: 919-844-7510

New York, NY
Tel: 631-435-6000

San Jose, CA
Tel: 408-735-9110
Tel: 408-436-4270

Canada - Toronto
Tel: 905-695-1980
Fax: 905-695-2078

ASIA/PACIFIC

Australia - Sydney
Tel: 61-2-9868-6733

China - Beijing
Tel: 86-10-8569-7000

China - Chengdu
Tel: 86-28-8665-5511

China - Chongqing
Tel: 86-23-8980-9588

China - Dongguan
Tel: 86-769-8702-9880

China - Guangzhou
Tel: 86-20-8755-8029

China - Hangzhou
Tel: 86-571-8792-8115

China - Hong Kong SAR
Tel: 852-2943-5100

China - Nanjing
Tel: 86-25-8473-2460

China - Qingdao
Tel: 86-532-8502-7355

China - Shanghai
Tel: 86-21-3326-8000

China - Shenyang
Tel: 86-24-2334-2829

China - Shenzhen
Tel: 86-755-8864-2200

China - Suzhou
Tel: 86-186-6233-1526

China - Wuhan
Tel: 86-27-5980-5300

China - Xian
Tel: 86-29-8833-7252

China - Xiamen
Tel: 86-592-2388138

China - Zhuhai
Tel: 86-756-3210040

ASIA/PACIFIC

India - Bangalore
Tel: 91-80-3090-4444

India - New Delhi
Tel: 91-11-4160-8631

India - Pune
Tel: 91-20-4121-0141

Japan - Osaka
Tel: 81-6-6152-7160

Japan - Tokyo
Tel: 81-3-6880-3770

Korea - Daegu
Tel: 82-53-744-4301

Korea - Seoul
Tel: 82-2-554-7200

Malaysia - Kuala Lumpur
Tel: 60-3-7651-7906

Malaysia - Penang
Tel: 60-4-227-8870

Philippines - Manila
Tel: 63-2-634-9065

Singapore
Tel: 65-6334-8870

Taiwan - Hsin Chu
Tel: 886-3-577-8366

Taiwan - Kaohsiung
Tel: 886-7-213-7830

Taiwan - Taipei
Tel: 886-2-2508-8600

Thailand - Bangkok
Tel: 66-2-694-1351

Vietnam - Ho Chi Minh
Tel: 84-28-5448-2100

EUROPE

Austria - Wels
Tel: 43-7242-2244-39
Fax: 43-7242-2244-393

Denmark - Copenhagen
Tel: 45-4450-2828
Fax: 45-4485-2829

Finland - Espoo
Tel: 358-9-4520-820

France - Paris
Tel: 33-1-69-53-63-20
Fax: 33-1-69-30-90-79

Germany - Garching
Tel: 49-8931-9700

Germany - Haan
Tel: 49-2129-3766400

Germany - Heilbronn
Tel: 49-7131-72400

Germany - Karlsruhe
Tel: 49-721-625370

Germany - Munich
Tel: 49-89-627-144-0
Fax: 49-89-627-144-44

Germany - Rosenheim
Tel: 49-8031-354-560

Israel - Ra'anana
Tel: 972-9-744-7705

Italy - Milan
Tel: 39-0331-742611
Fax: 39-0331-466781

Italy - Padova
Tel: 39-049-7625286

Netherlands - Drunen
Tel: 31-416-690399
Fax: 31-416-690340

Norway - Trondheim
Tel: 47-7288-4388

Poland - Warsaw
Tel: 48-22-3325737

Romania - Bucharest
Tel: 40-21-407-87-50

Spain - Madrid
Tel: 34-91-708-08-90
Fax: 34-91-708-08-91

Sweden - Gothenberg
Tel: 46-31-704-60-40

Sweden - Stockholm
Tel: 46-8-5090-4654

UK - Wokingham
Tel: 44-118-921-5800
Fax: 44-118-921-5820